

Bill Bridge: A Written Summary

Why Bill Bridge

A bill is a proposed law, not a law yet. In Kenya, it starts with drafting, usually by a ministry working with the Attorney-General's Chambers, goes to Cabinet for approval, then gets published in the Kenya Gazette so the public can see it [1, 2]. From there it moves through Parliament in stages: First Reading, Second Reading (where the general debate and, in practice, most public participation happens), Committee of the Whole House, Third Reading, and finally Presidential Assent [3, 4].

There exists a gap that this project aims to fill. The legal window for a citizen to weigh in on a bill sits early, mostly around the committee and second-reading stages, before the bill has been fully shaped and voted on. But actual public awareness, in practice, tends to show up much later. Often right before or right after a bill passes. Often triggered by something going viral, not by anyone having read the bill itself.

The clearest recent example is the 2024 Finance Bill(Kenya). One person who took part in the protests described it plainly: they'd "heard about the bill in passing" for weeks but didn't pay real attention until a tweet, and the phone numbers of MPs it included, made the moment feel urgent [5],an occurrence that would happen again. Afrobarometer's pre-protest survey data shows mobilisation in the run-up to the 2024 protests ran mostly through X, TikTok, and Instagram, where the bill got simplified for public consumption after the fact, not explained ahead of time through an official channel [6]. The bill had already passed its first vote, 195 to 106, by the time mass protests broke out [7]. And when Kenya's Court of Appeal later reviewed the whole episode, it found the Finance Bill unconstitutional specifically for insufficient public participation [8].

So the pattern looks like this: a narrow, early, legally-mandated window to comment, a long quiet stretch where almost nobody outside Nairobi civil society is paying attention, then a sudden, reactive, often anger-driven spike in awareness once a bill has already advanced or passed. People do "chip in." Just usually too late to shape the bill, and usually without having read what it actually says.

Bill Bridge tries to move that engagement earlier. Plain language, in the language people actually use, delivered at the stage a bill is introduced, not after it's already a headline.

Existing tools, and why they don't fully close this gap

It is important to acknowledge that this isn't an empty space. A few real tools already do parts of this work, and it's worth naming them.

Mzalendo Trust has tracked Kenya's Parliament since 2005 and runs Dokeza, a platform where citizens can annotate specific clauses of a bill and submit memoranda directly [9, 10]. It's a genuinely

strong tool. But Mzalendo’s own published research names the exact problem this project is trying to address: digital participation is “neither widespread nor equitable,” held back by data costs, limited digital literacy, and low awareness of the tools even existing [9]. A separate case study of Kenya’s legislative accountability landscape put it more realistically: low public demand for this kind of information, paired with limited access to it, are the two things actually holding these tools back [11].

Twitter/X is where the Finance Bill conversation actually happened in 2024, and it moved fast and reached a huge number of people [6, 7]. But a platform built for velocity isn’t built for reference. A thread from three weeks ago about a bill’s actual clauses is functionally gone, buried under everything posted since, and there’s no way to go back and find what a bill actually says once the moment has passed.

Nigeria’s National Assembly runs a comparable official bill tracker, BillTracker, maintained by the National Assembly Library Trust Fund [12, 13]. It’s a legitimate model for what an official, government-run tracker can look like. But like most web-based trackers, it assumes the citizen already knows to go looking for it, has the data to load it, and has the literacy in legal language to make sense of what they find.

Bill Bridge doesn’t replace any of these. It tries to sit earlier in the chain: reachable without a search, without a data bundle, and without needing to already understand how a bill is written, so that by the time someone does encounter Mzalendo, or a viral thread, or the actual gazette, they’re not starting from zero.

Why SMS and USSD specifically

The honest version of this argument isn’t “everyone reads their texts and ignores everything else.” Current data doesn’t actually support that. SMS traffic in Kenya has been falling for several quarters running, as more people shift to WhatsApp and other internet-based messaging on increasingly cheap smartphones [14, 15, 16]. Smartphone penetration crossed 90% by the end of 2025 [16]. If the only question were what people prefer to read, SMS wouldn’t be the obvious answer anymore.

The real argument is about reach, not preference. Kenya has more active mobile connections than people, somewhere around 145 to 150 percent penetration depending on the quarter measured [14, 15]. But internet penetration sits at roughly 40.5%, and social media usage at about 31.8% of the population [17]. That gap, everyone with a phone number against a much smaller group reliably online, is exactly the gap Mzalendo’s own research names as a real barrier: high data costs specifically keep people out of digital civic participation [9]. SMS and USSD don’t need a data bundle. They work on a basic phone as well as a smartphone. They reach the roughly 60% of Kenyans who a web app, however well designed, simply won’t reach on a given day. That’s the actual case for this channel: not that it’s preferred, but that it’s the widest net available for something meant to reach the common mwananchi, not just someone already online and already engaged.

Track

Primary track: Transparency & Accountability. Bill Bridge makes one specific kind of government decision visible and understandable in real time: bills before Parliament, before they become law, not after.

Secondary connection to Stability & Social Cohesion. The project’s anonymous, no-signup web discussion feature gives citizens a shared, low-friction space to talk about a specific bill with others who’ve read the same plain-language summary. A small step toward the kind of trusted, grounded communication that track describes.

Information sources

- **Kenya Law** (new.kenyalaw.org): the primary source for bill text itself. Bills are published as PDFs, rendered through a JavaScript PDF viewer rather than plain HTML.
- **Mzalendo Trust’s Bill Tracker:** a best-effort source for legislative stage and sponsor information, where Kenya Law’s own pages fall short.
- **Parliament of Kenya’s own Bill Digests:** used where a bill’s official digest was more complete than the gazetted text, including real public-participation deadlines and submission addresses.

Both Kenya Law and Mzalendo reuse bill titles across different years. There are three separate “Public Participation Bill” entries alone (2023, 2024, 2025). Matching across sources has to key on title plus year together, never title alone, or it will silently attach the wrong bill’s metadata to the wrong text. We ran into this directly, and designed around it.

What’s automated versus manually curated, honestly: the decode, translate, notify, and question-answering pipeline is fully automated and works on any bill text fed into it. Sourcing that text is only partly automated for this proof of concept. `extract_one_bill.py` is a real, working script, not a mockup. It finds a bill on Kenya Law’s live listing by link text, correctly identifies that bills render through a JS-loaded PDF viewer (the real PDF URL lives in a `data-pdf` attribute found by inspecting an actual fetched page, not guessed), downloads it, and extracts the text. Run against a real bill, it pulled 63,753 characters matching a second, independently-sourced copy of the same bill. It does not yet feed that extraction automatically into the decode pipeline, or run on a schedule. Both are deliberate scope decisions for a two-day build. The three bills in the working prototype were sourced and verified by hand.

Approach to trust and accuracy

Trust was a design constraint from day one.

- **Every claim in a decoded summary cites its source.** Each explanation pairs with a `source_ref` (say, “Section 14(3)”) and the actual wording it’s explaining, so anyone can check a claim against the real bill, inside the app.
- **The bill’s actual wording never gets translated.** In Kiswahili, the explanation is translated, but the quoted `official_text` stays in English, because that’s how the bill is officially gazetted. Translating a legal quote would mean showing a translation of a quote, not the quote itself.
- **The model is explicitly told never to invent facts.** Where a bill states no specific public-comment deadline, true of the actual National Coroners Service Bill we tested against, the decode correctly left that field empty instead of defaulting to a pattern seen elsewhere. We checked this by hand, decoded output against full source text, for every bill used in the demo.
- **A real translation error got caught and fixed through testing.** “Wakorona” (coroner) is genuine, established Kiswahili legal terminology [18], but risks being misread as a COVID-19

reference without context, which can happen in a bare USSD menu. The translation prompt now adds a short clarifying phrase the first time the term shows up in a title, and leaves it alone elsewhere, where context already makes the meaning clear.

- **Gaps get disclosed, not hidden.** When Mzalendo has no record of a very recently gazetted bill, the app falls back to the bill’s own memorandum for sponsor information. The support/oppose feature is labeled in the app as a simple tally, not a formal petition. Comments are labeled as lightly filtered, not human-moderated.

How AI tools were used

AI was used in two genuinely different ways here.

AI as a core product feature. The decode, translation, and SMS question-answering pipeline runs on Google’s Gemini API, live, inside the application. This is the actual mechanism that makes plain-language, bilingual, cited civic information possible at all.

AI (Claude) as a development and research partner, across the whole build:

- **Architecture and scoping,** before writing any code: what to automate first, what to deliberately leave manual for a two-day build, and why.
- **Code generation across the full stack:** the FastAPI backend, USSD menu logic, SMS integration, database schema, no-signup web frontend.
- **Live web research that actually changed the build.** Mzalendo Trust on recommendation was used as a data source, verified Kenya Law’s real page structure from actual fetched HTML instead of guessing at selectors, confirmed real Kiswahili legal usage of “Korona”/“Wakorona” against real Kiswahili-language legal press, and pulled the citizen-engagement research behind the “Why Bill Bridge” section above.
- **Real, iterative debugging.** An `SSL: WRONG_VERSION_NUMBER` TLS failure connecting to Africa’s Talking’s API, on this specific Windows machine, even though curl and browsers connected fine. It took four genuine attempts to diagnose: a legacy-renegotiation theory, ruled out by direct testing; a TLS-version-cap theory, also ruled out; before landing on a working fallback that shells out to curl when Python’s own TLS stack fails. Each attempt got tested before moving to the next one.
- **Prompt engineering as an explicit, documented part of the system.** The decode prompt’s no-invention rule, the official-text-stays-English rule, and the Wakorona disambiguation rule are deliberate, written instructions, refined through testing against real bills.

The idea for Bill Bridge, its scope, and every product decision in it were the author’s own. AI was used to build it, research it, and debug it. Not to come up with it.

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